POLITECNICO MILANO 1863 | DIPARTIMENTO DI INGEGNERIA CIVILE E AMBIENTALE

Fast Urban Digital Twin prototyping based on open data



Juan Pablo
Duque Ordoñez

FOSS4G AUCKLAND 2025

GEOLAB Geomatics and Earth Observation lab

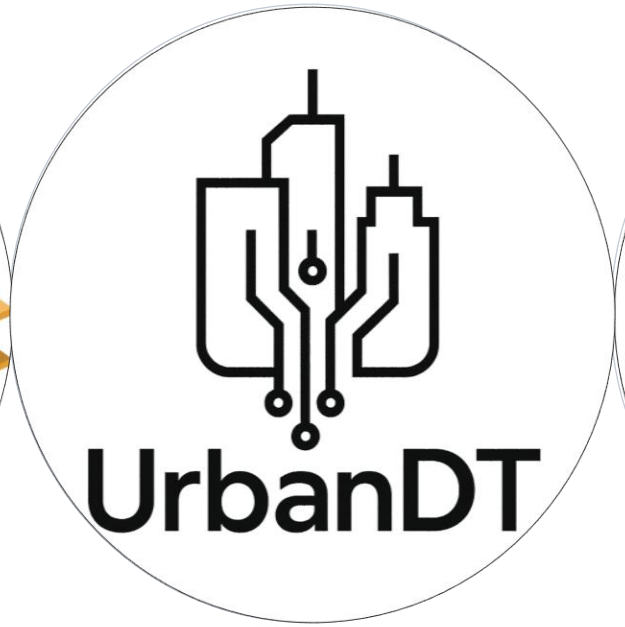
BUZZWORD ALERT!

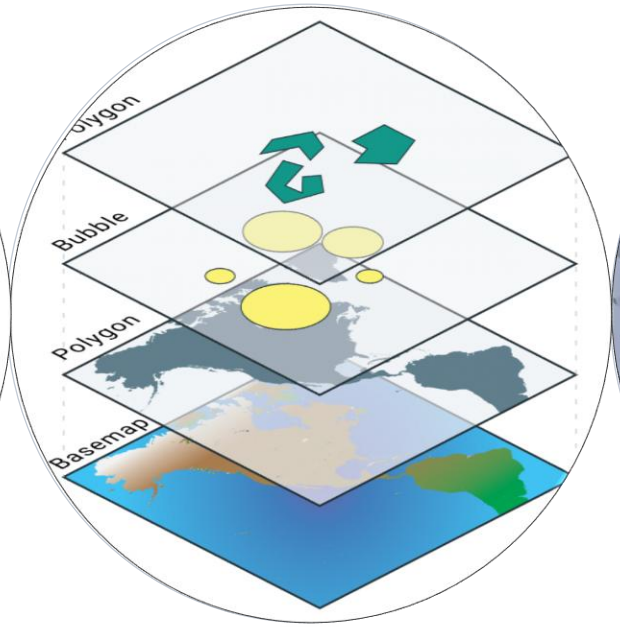
Urban Digital Twin(s)

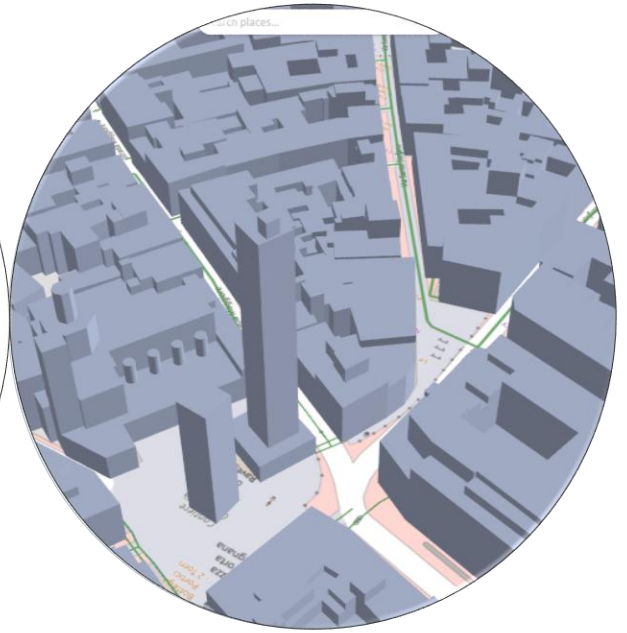
Overview

Context > Urban Digital Twin Prototyping Platform > Data Requirements and Automated Data Download > Use Case









Context

01

Context – Definition

But first...

What is actually an Urban Digital Twin?

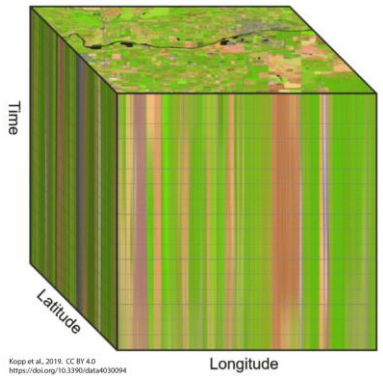
*"A City/Urban Digital Twin is a **spatial–temporal virtual representation** of a real-world urban area or city, **mirroring its states** during its lifecycle through IoT sensors. It is used to **monitor and analyse** urban systems across different time spans to **aid in decision-making** and can be extended to **simulate and predict** various states and scenarios." **

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Usage of Geospatial and Temporal Data

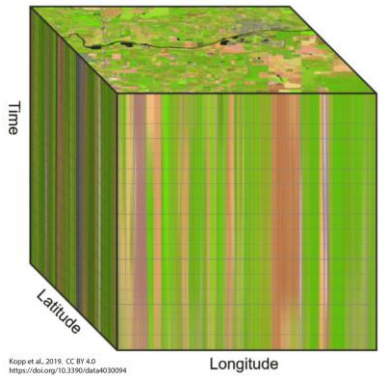


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Usage of Geospatial and Temporal Data



Constant Monitoring



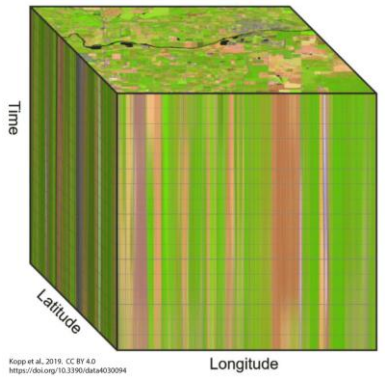


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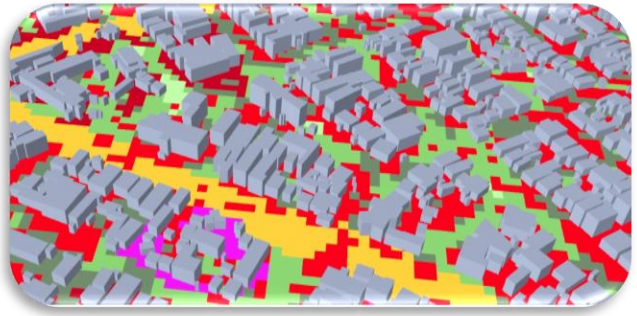
Usage of Geospatial and Temporal Data





Constant Monitoring





Analyses and Simulations

* Mahmoud Abdelrahman et al. "What is a Digital Twin anyway? Deriving the definition for the built environment from over 15,000 scientific publications". https://www.sciencedirect.com/science/article/pii/S0360132325002306.

Context – Research Objectives

Contextualise, document, and implement a software platform for the creation of Urban Digital Twin (UDT) prototypes – This to assemble **minimal, globally replicable** UDT implementations, fostering their adoption as decision making tools.

1. Identify system components to propose a system architecture for the implementation of UDT systems.

2. Identify key datasets for UDTs and define specific global open datasets that provide such data.

3. Develop an integrated software platform for UDT implementation.

4. Formulate an automated pipeline for provisioning minimal data based to UDT prototypes.

5. Implement an Urban Digital Twin for testing and validation.

Context – System Components and Data

Systematic literature review to identify the UDT state-of-the-art from a **technological** perspective.

System Components

Data Requirements

Context – System Components and Data

Some key components for Urban Digital Twins:

- 👀 Visualisation
- ⚙️ Processing
- 📊 Data Management

Data requirements:

- 🏢 3D Data is fundamental For UDTs.
- 🚗 Street networks are common data requirements.
- 👌 Land cover, population / demographics, elevation models used to enhance.
- 🌐 IoT is widely used for continious monitoring.

Urban Digital Twin Prototyping Platform

02

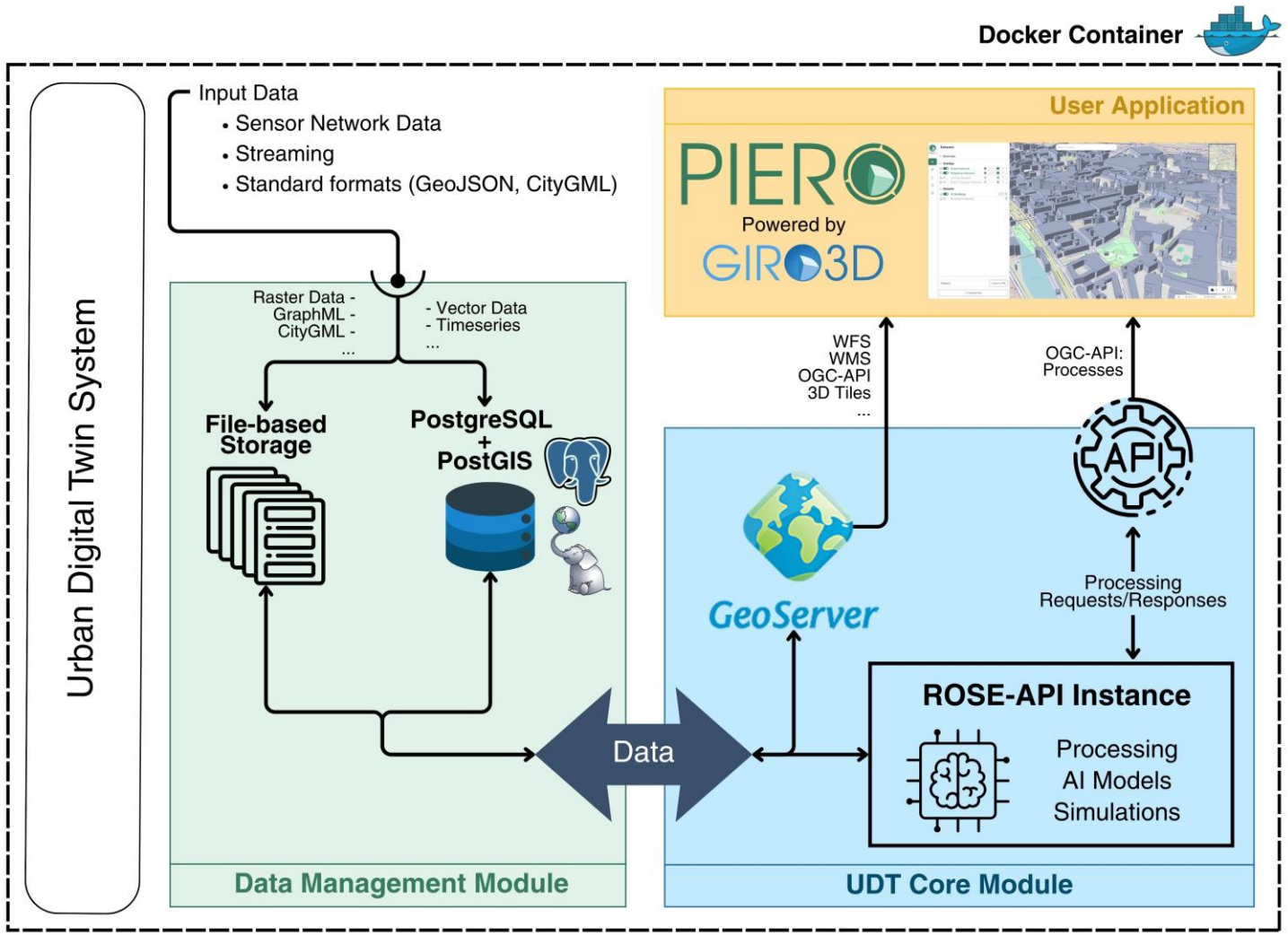
Platform Overview – System Structure

UrbanDT

Starting from the identified system components we developed UrbanDT!

- Integrated software platform for hosting UDT implementations.
- Supports 3D visualisation, custom processing, multiple data formats, and data access through OGC Web Services and OGC APIs.
- Based entirely on open source software.

Platform Overview – System Structure



System Architecture

Three-module architecture:

- **Data Management Module**

File-based storage for raw data and geospatial database for structured data.

- **Core Module**

ROSE-API and GeoServer provide custom processing an data sharing capabilities.

- **User Application Module**

User interaction and (2D/3D) visualisation.

Platform Overview – Core Module

- **ROSE-API** web server for configuration management and processing capabilities.

- **GeoServer** for data sharing and data integration through OGC standards.

- Processing capabilities through standardised **OGC API** interfaces.
- Implements **OGC API: Processes** for custom processing capabilities.
- **Open-source software** and implemented in the context of the research.

- Implements OGC web services for data sharing (WMS, WFS, WCS)



GeoServer

Platform Overview – Data Management Module

- **PostgreSQL and PostGIS geospatial database** for vector data handling.





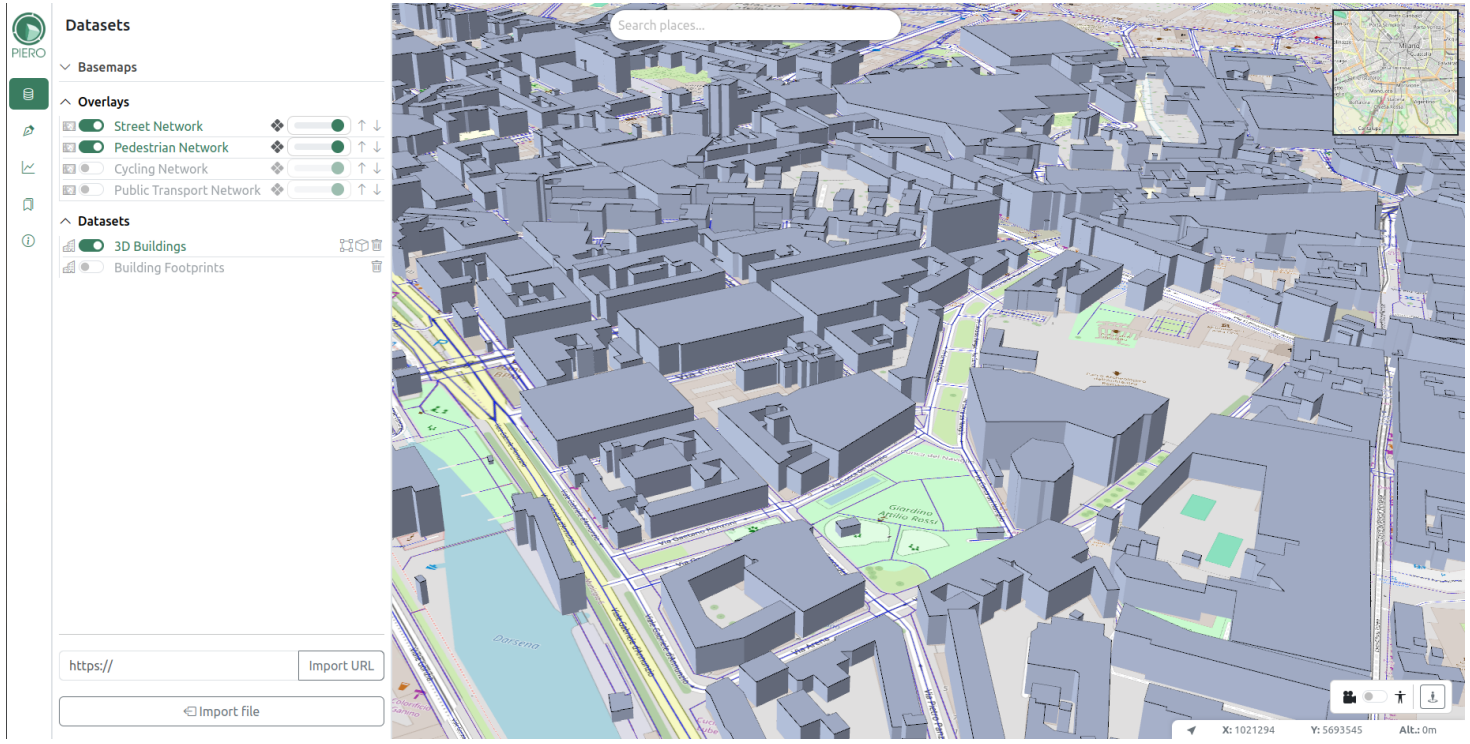
- Handling of vector geometries.
- Connected with GeoServer for exposing data.

- **File-based** storage for raster and raw data management.

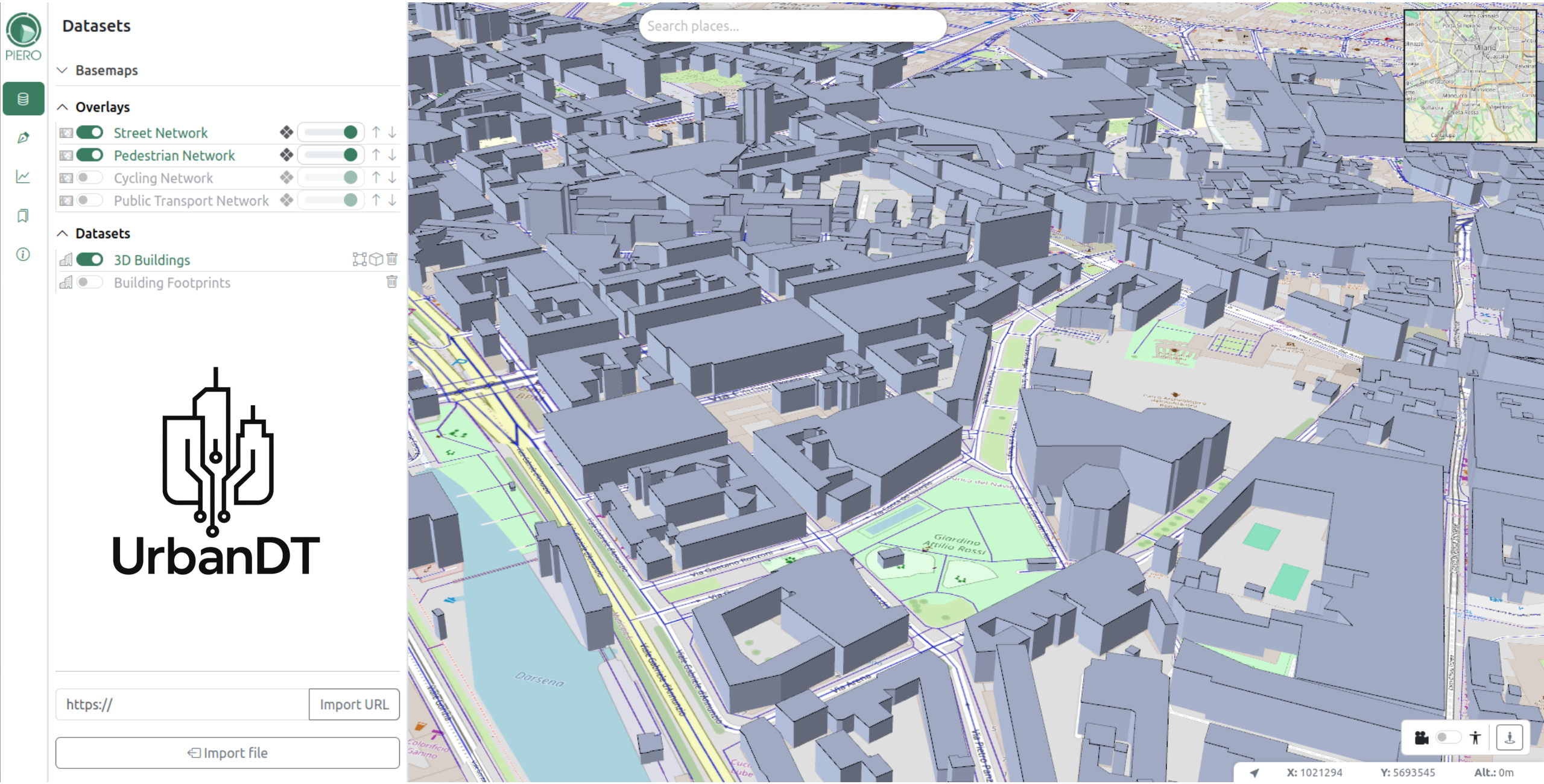
- **Metadata-based** mapping for data discovery.
- **Raw file storage** for analysis and handling of custom data formats.
- Handling of **raster** data formats.

Platform Overview – User Application Module

- **Piero-based** user application, which is an open source digital mapping application.
- Based on **Giro3D**, an open-source 3D web engine.



- Supports **2D** and **3D** datasets and visualisation.
- Handles data using **OGC standards** (WMS, WFS, GeoJSON, CityGML, etc).
- **Dynamic configuration**, enabling data adding data on-the-fly.


PIERO
Datasets
Basemaps
Overlays
Street Network
Pedestrian Network
Cycling Network
Public Transport Network
Datasets
3D Buildings
Building Footprints
UrbanDT
https://
Import URL
Import file
Search places...
Darsena
Giardino Attilio Rossi
X: 1021294
Y: 5693545
Alt.: 0m

Automated Data Download

03

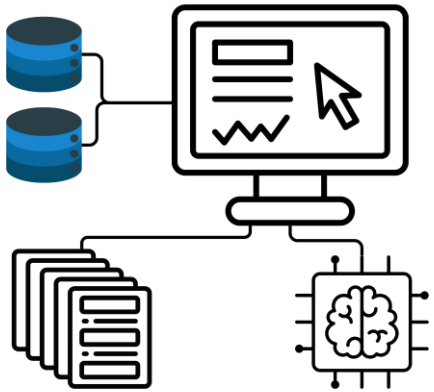
~~Automated Data Download~~

Assembling Urban Digital Twin Prototypes

03

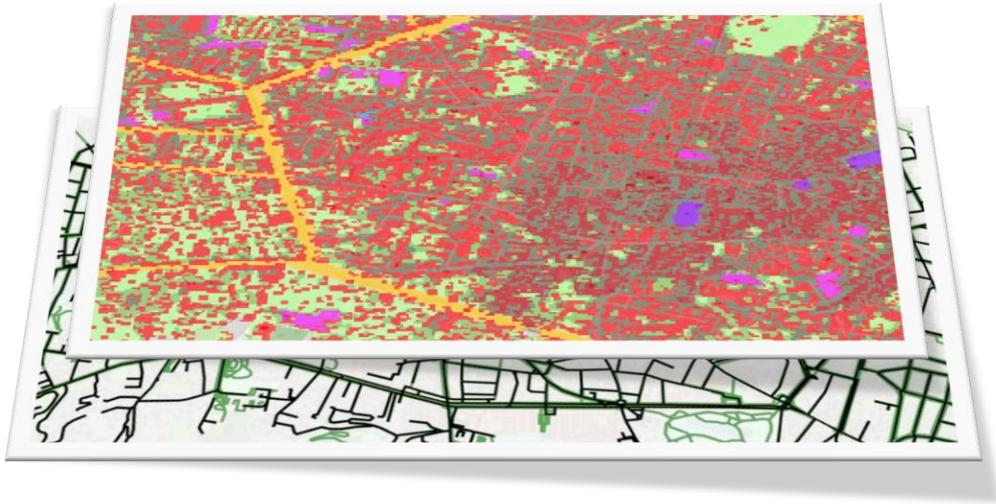
Urban Digital Twin Prototype

Urban Digital Twin System



+

Base Data



*All based on FOSS

*All based on free global open data

Urban Digital Twin Prototypes – Base data requirements for UDTs

3D City Model

3D-GloBFP

Microsoft Global ML Building Footprints

3D city model as building footprint extrusions.

Street Network

OpenStreetMap

Pedestrian
Cycling
Road
Public Transport

Digital Elevation Model

NASADEM Dataset

Global coverage
30m resolution

Land Cover

Copernicus CGLS-LC100

Google Dynamic World v1

GHS BUILT-C

Three datasets representing general, dynamic, and urban landcover.

Population

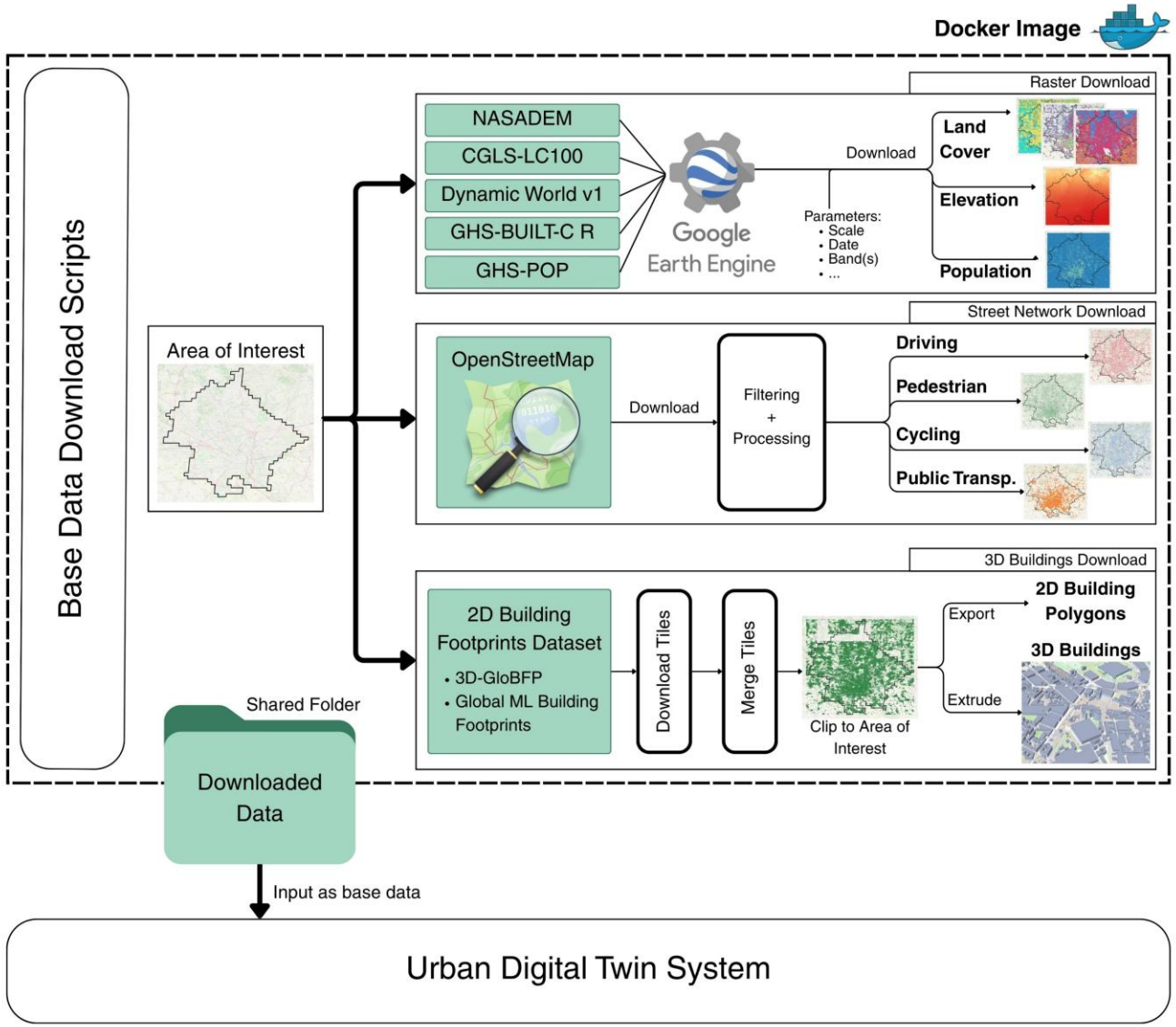
Global Human Settlement GHS-POP R2023A

Multitemporal
100m resolution
population grid

Urban Digital Twin Prototypes – Automated Download Pipeline

- **Automated pipeline** to download and process base UDT data.
- The data is used as **default** data for **UDT prototypes**.
- **Raster data** (land cover, digital elevation model, and population) leverages **Google Earth Engine*** for its download.

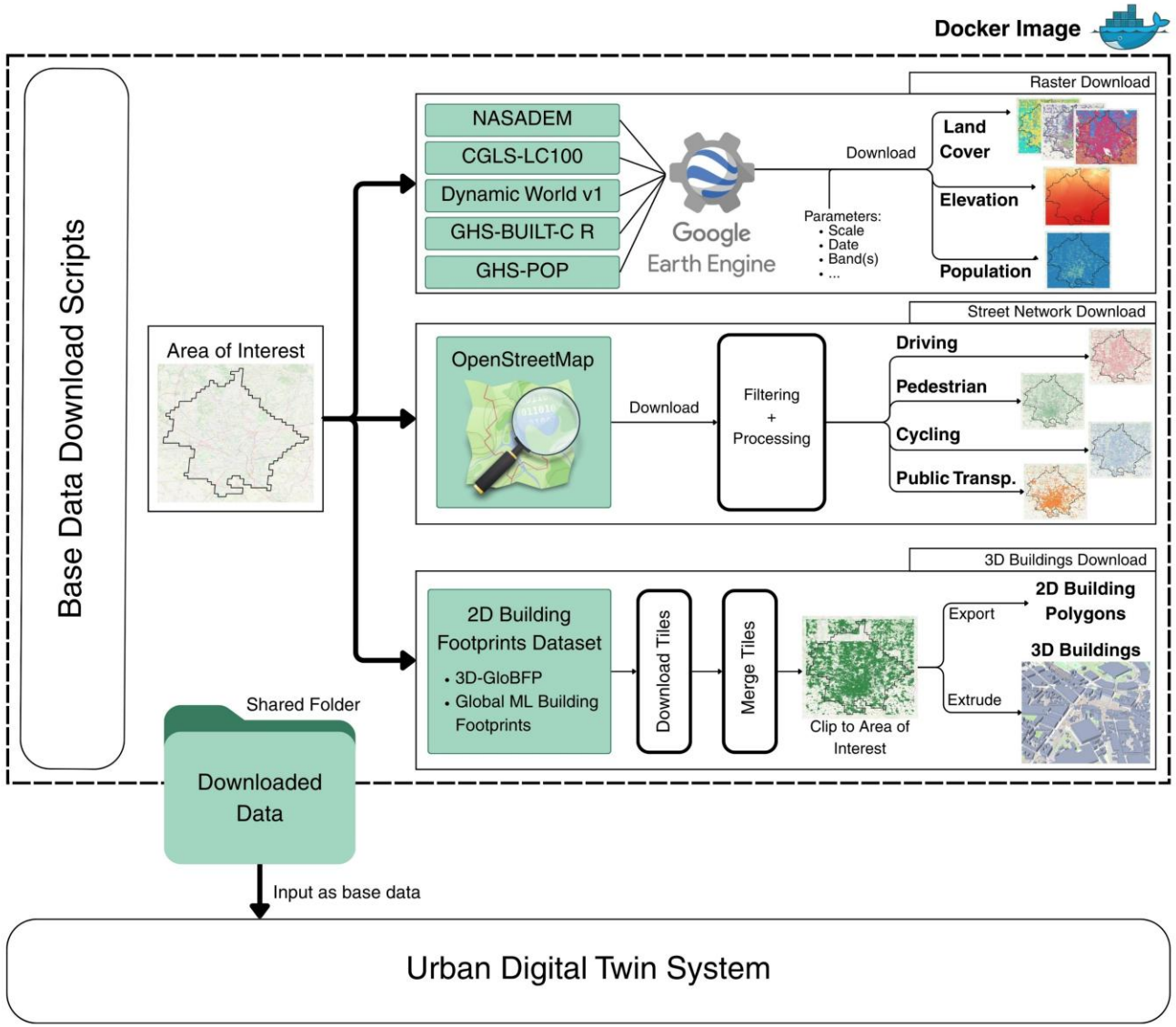
Not particularly open-source, but 🤫



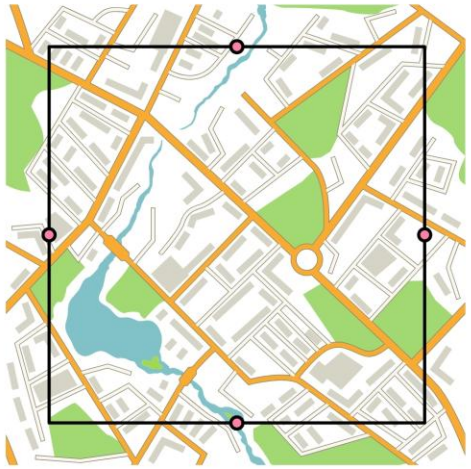
Urban Digital Twin Prototypes – Automated Download Pipeline

- **Custom filters and algorithms** are used to build **realistic** pedestrian, cycling, road, and public transport street networks from **OpenStreetMap**.

- Global 2D **building footprint** data and building heigth information is used to build a **3D city model** from building footprint **extrusions**.



Urban Digital Twin Prototypes – Creating the UDT



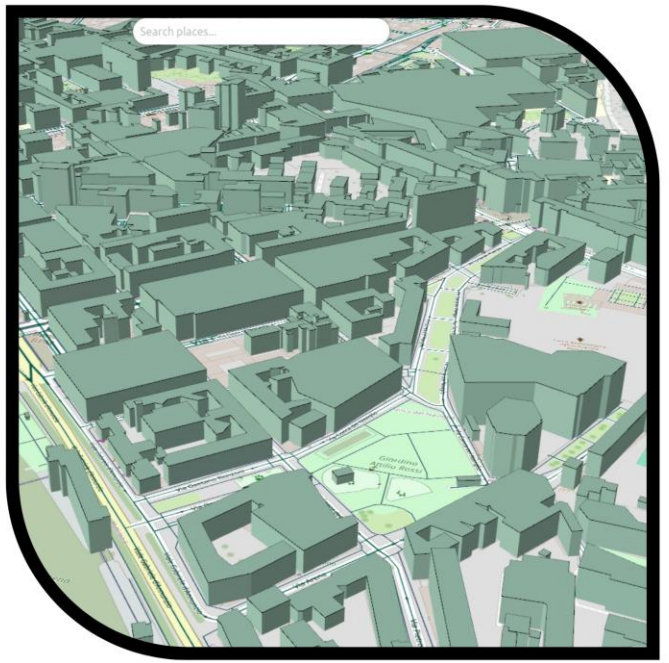
Map Extent

And some other configuration

\+

UrbanDT

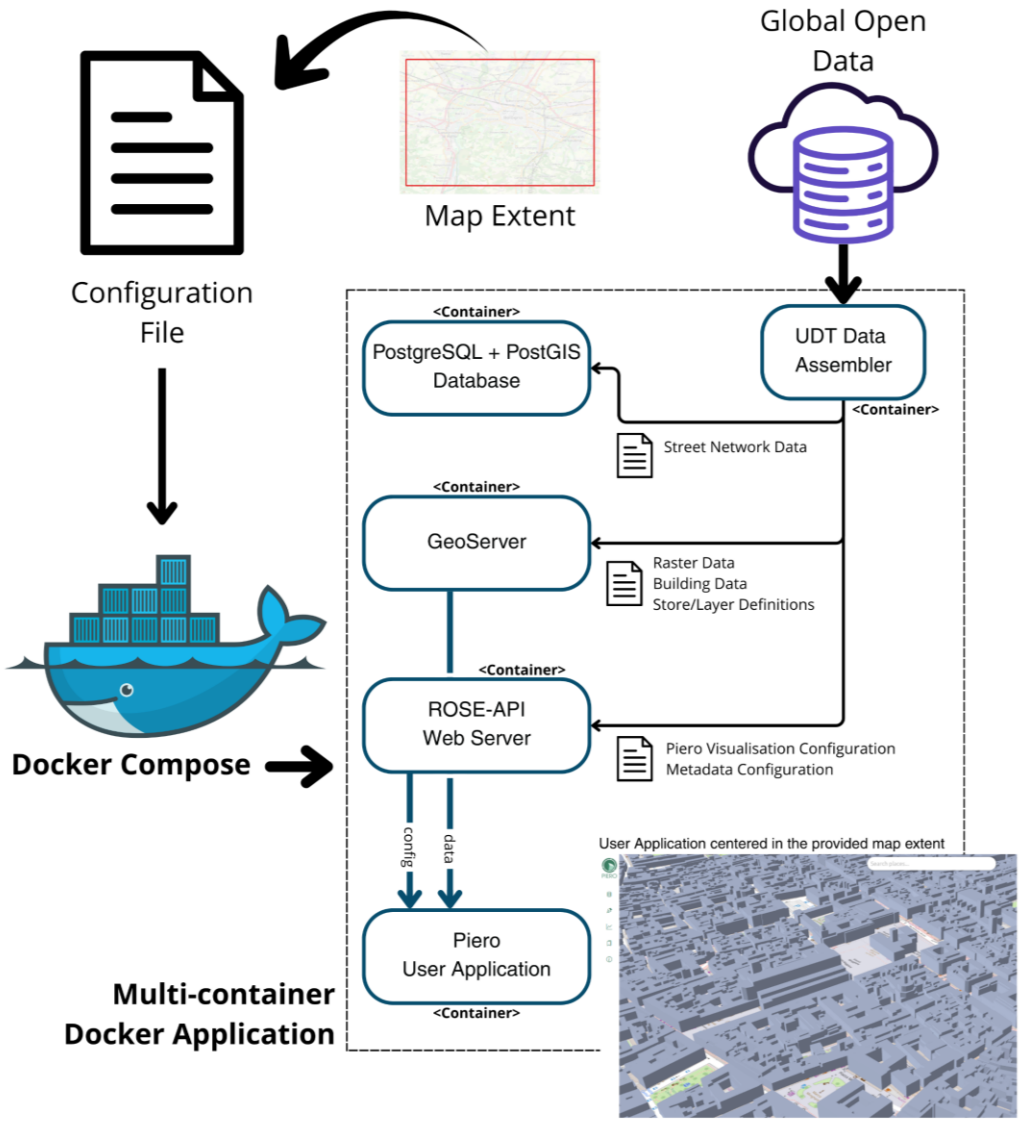
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Urban Digital Twin Prototypes – Creating the UDT

An automated pipeline generates the software infrastructure and inserts **base data** to create an on-demand Urban Digital Twin from an area of interest.

- A **configuration file** is used to **customise.**
- A bounding box or area of interest **delimits** the boundaries for data download.
- **Docker** is utilised to **orchestrate** the system components and infrastructure.
- **Global open data** is downloaded and stored as base data.
- **User application configuration** is automatically generated and applied.



Use Case: Urban Digital Twin

04

Use Case: Urban Digital Twin

Demonstrating the global applicability of the proposed Urban Digital Twin prototyping framework, a case was implemented as a proof of concept to explore visualisation, processing & extension capabilities of the system in …



Use Case: Urban Digital Twin

Demonstrating the global applicability of the proposed Urban Digital Twin prototyping framework, a case was implemented as a proof of concept to explore visualisation, processing & extension capabilities of the system in

Auckland, New Zealand



Use Case: Urban Digital Twin Capabilities

Data Visualisation and Management



- 3D and 2D data visualisation capabilities.
- Flexible data insert and sharing.
- Custom styling through GeoServer.
- Standardised interfaces.

Processing and Simulations



- Supports custom processing scripts through ROSE-API
- Asynchronous, cloud-based processing using OGC API – Processes interfaces.
- Street network analysis and raster-based analyses offered by default

Constant Monitoring



- Still in process for constant monitoring capabilities.
- **Still theoretically possible by adding custom data!**

Use Case: Urban Digital Twin - Results

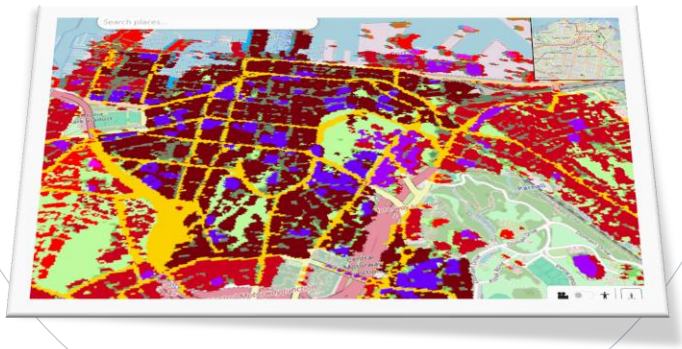
Data Exploration

Visual exploration of global open base data and integrated authoritative datasets.



Raster-based analysis

Raster-based population and land cover distribution analysis at the city and local scales.



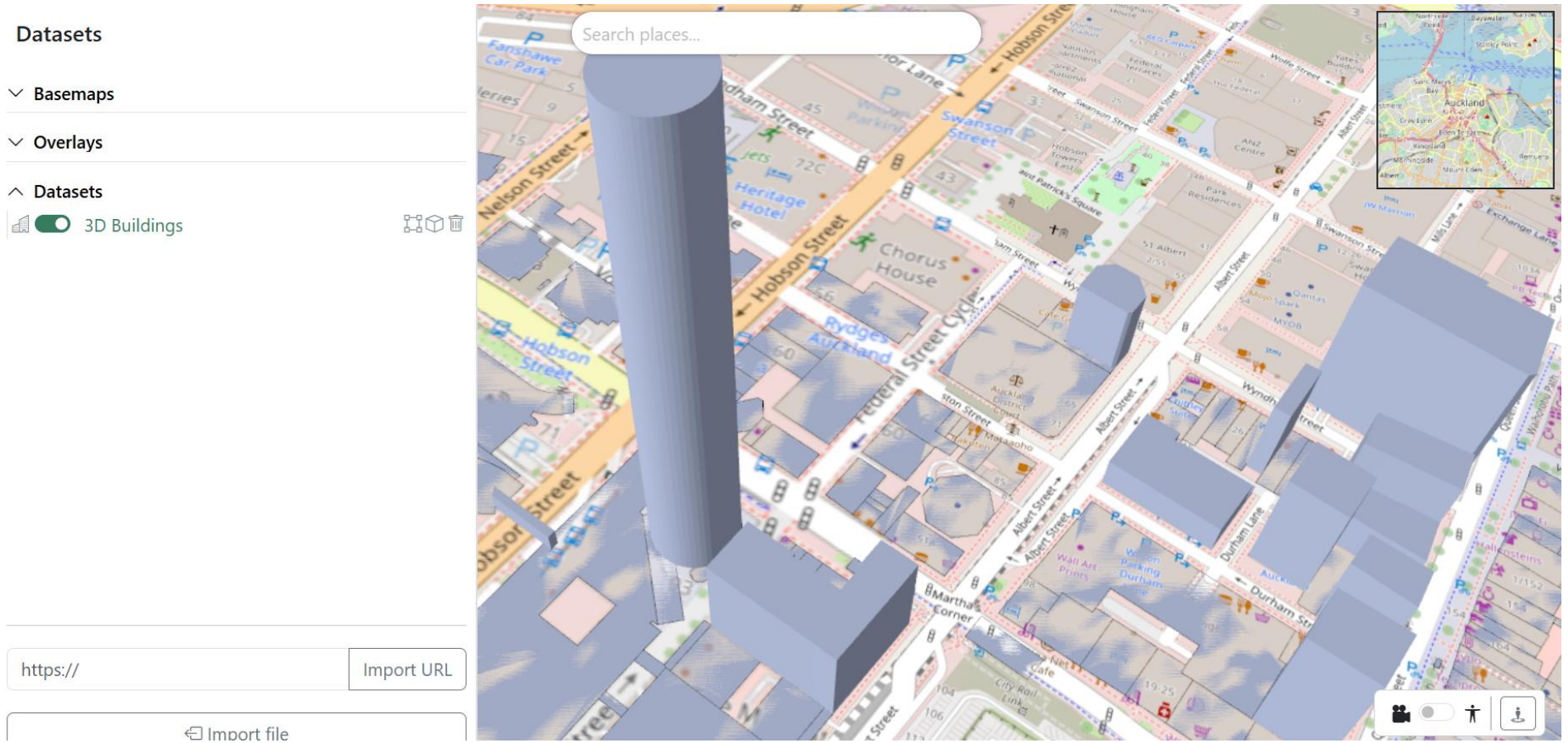
Street Network analysis

Calculation of various street network indices for assessing active and motorised mobility at the city and local scalse.



Use Case: Urban Digital Twin – Data Exploration

Building data not available in global datasets for New Zealand!
Custom Data from Overture downloaded and added to the UDT Prototype.



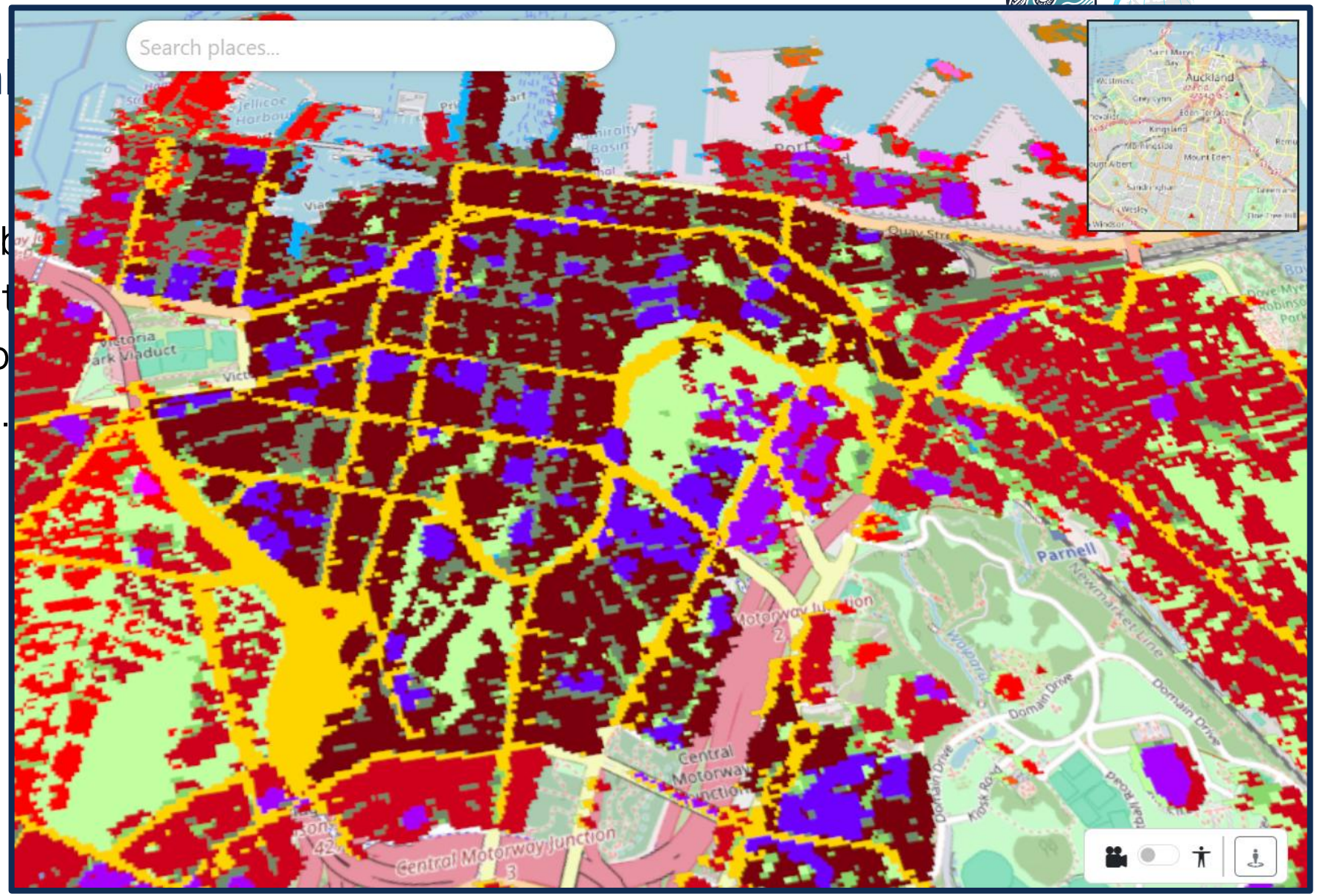
Use Case: Urban Mobility Digital Twin – Data Exploration

Raster visualisation capabilities:

- Various Land Cover datasets.
  - Digital Elevation Model.
    - Population Grid.

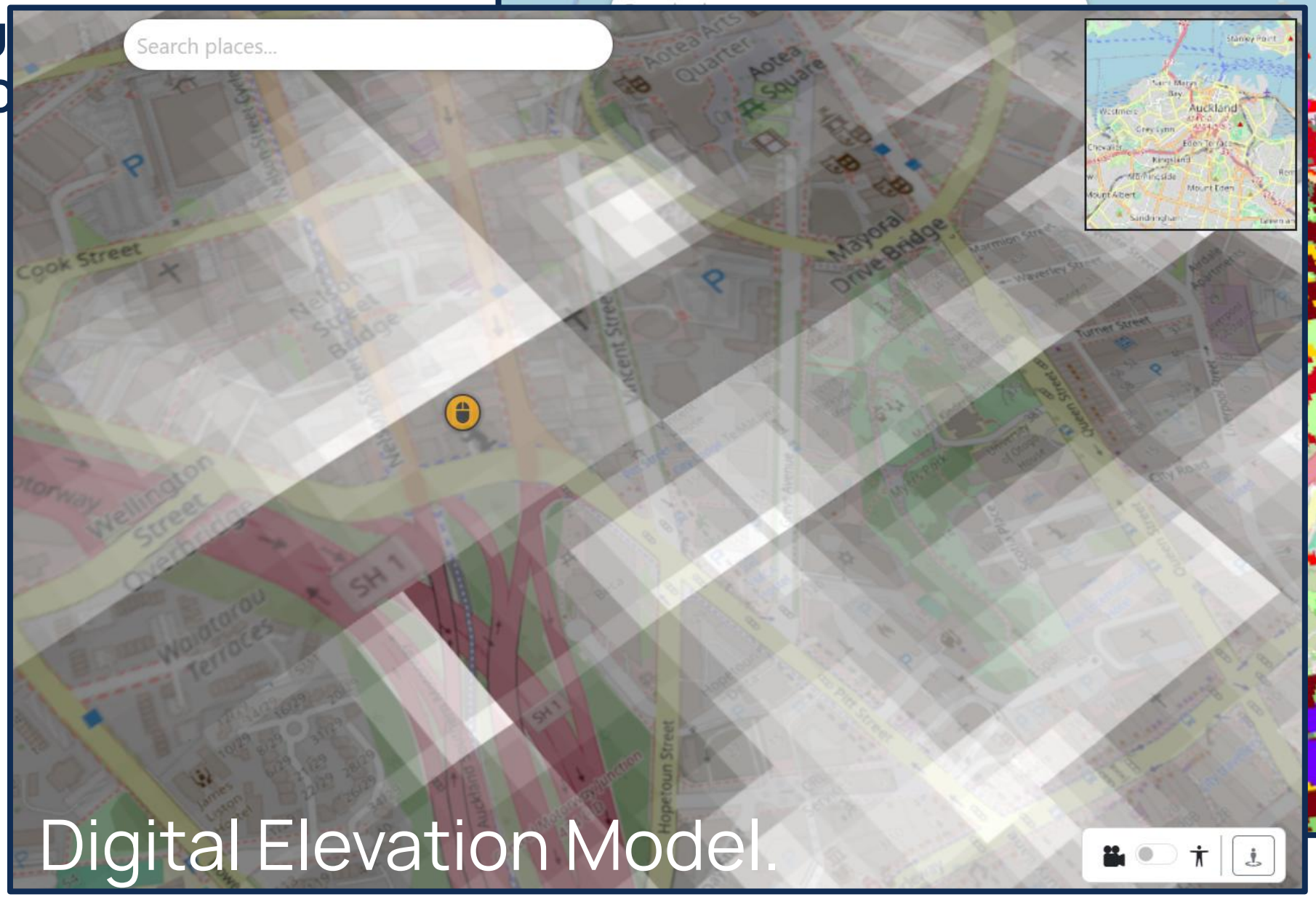
Use Case: Urban Digital Data Exploration

Raster visualisation capab
- Various Land Cover dat
  - Digital Elevation Mo
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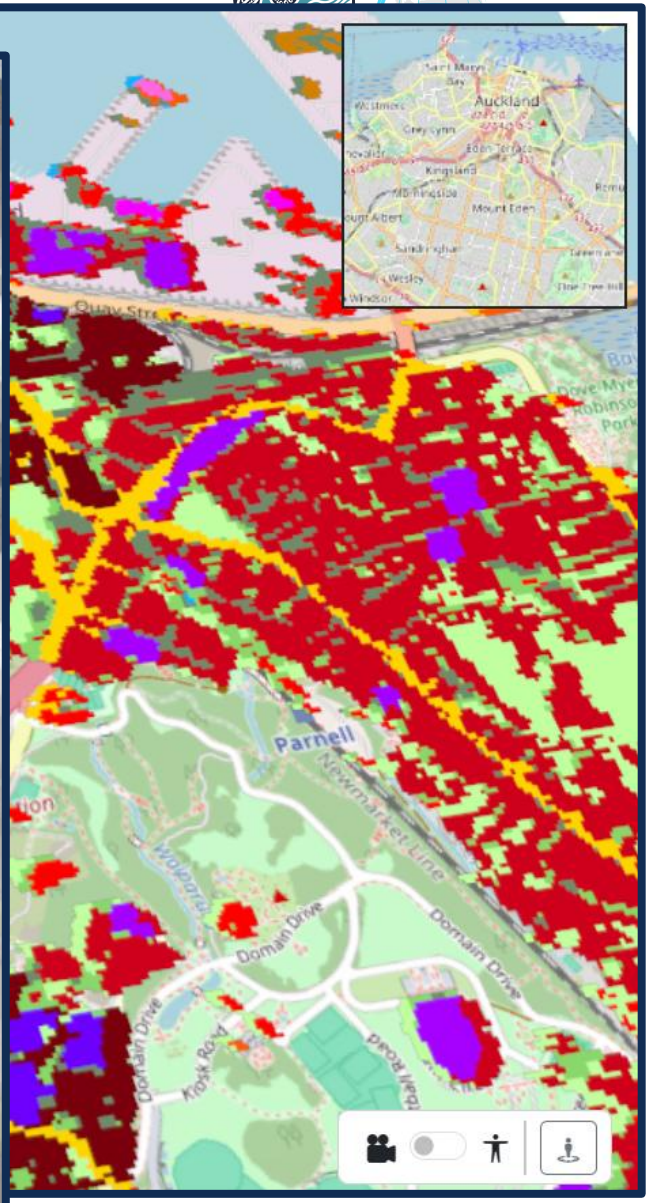


Urban Landcover

U
D



Digital Elevation Model.



Urban Landcover

Use Case: Urban Digital Twin – Data Exploration

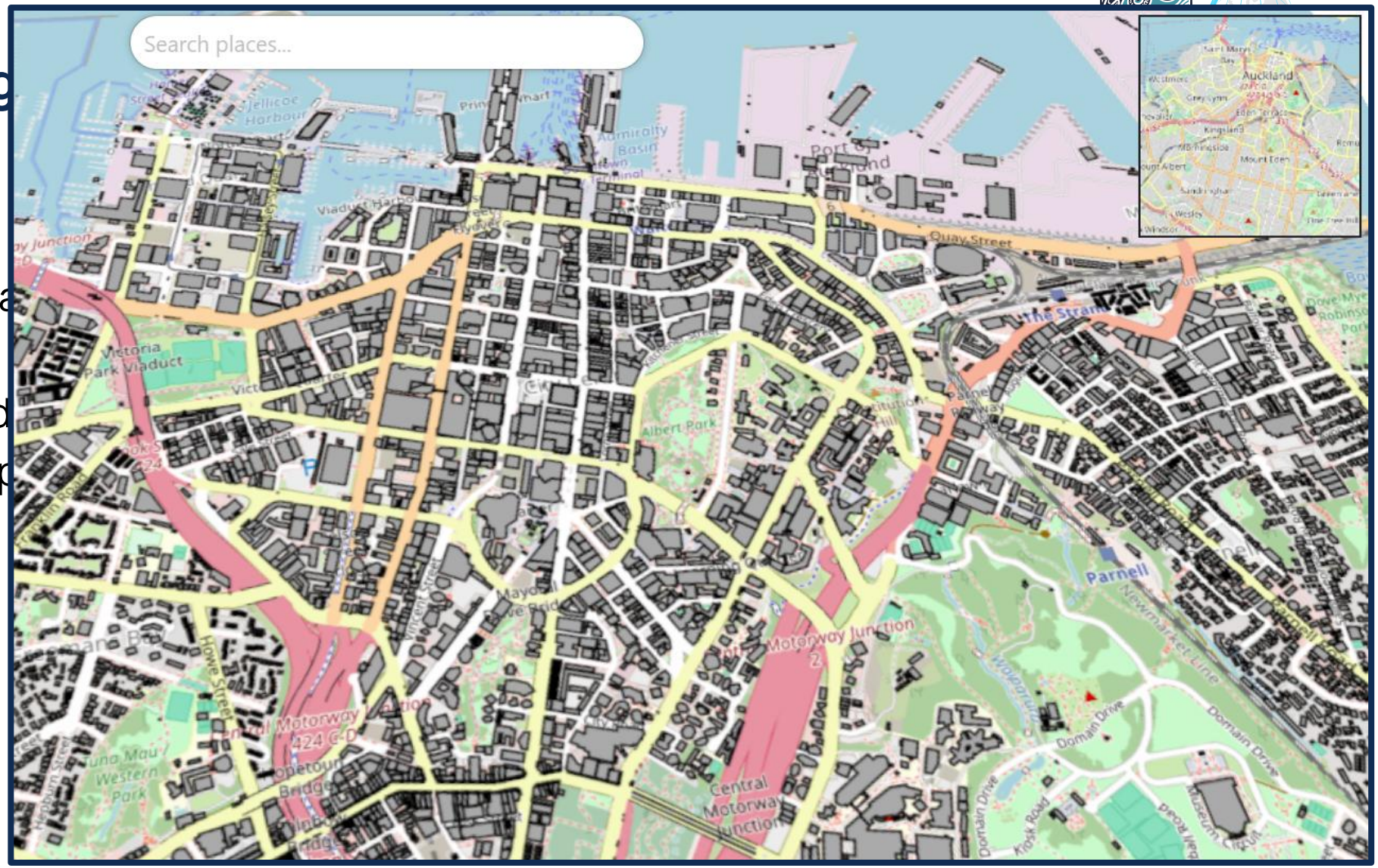
Vector visualisation capabilities:

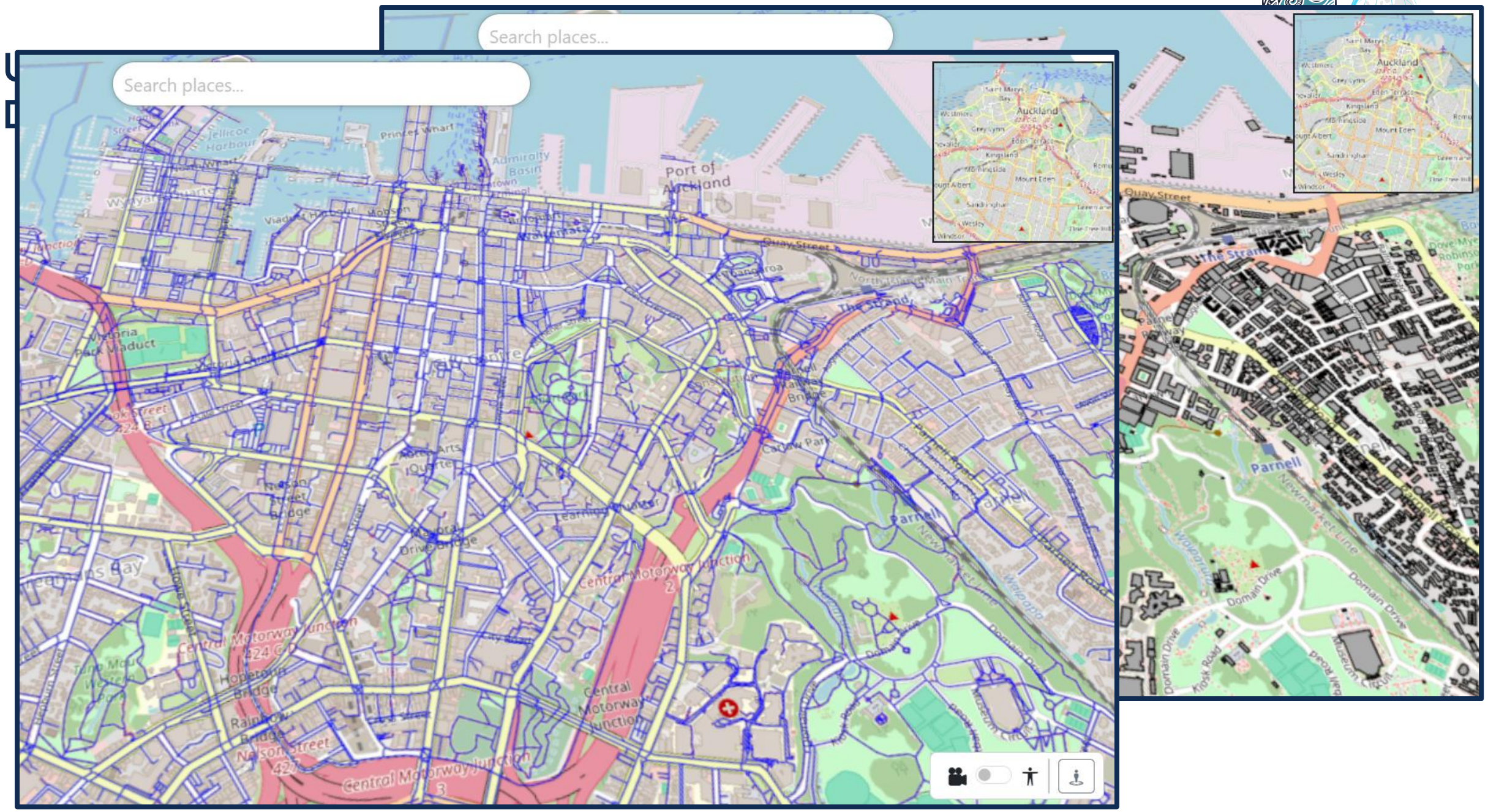
- Street Network datasets.
  - Building Footprints

Use Case: Urban Dig
Data Exploration

Vector visualisation ca

- Street Network d
  - Building Footp




Search places...
Search places...
Port of Auckland
Quay Street
The Strand
Parnell
Domain Drive
Central Motorway Junction
Victoria Park Viaduct
Hopetoun Bridge
Rainbow Bridge
Nelson Street
Auckland

Use Case: Urban Digital Twin – Analysis Results

Characterisation of **active and motorised mobility** through calculation of **street network indices**:

- Calculated for the **pedestrian, cycling, and road** networks.
- Proximity to public transport only calculated for pedestrian network, slope only for the cyclist, and orientation entropy only for the road.

Street Length

Proximity to Points of Interest

Link-Node Ratio

Proximity to Public Transport

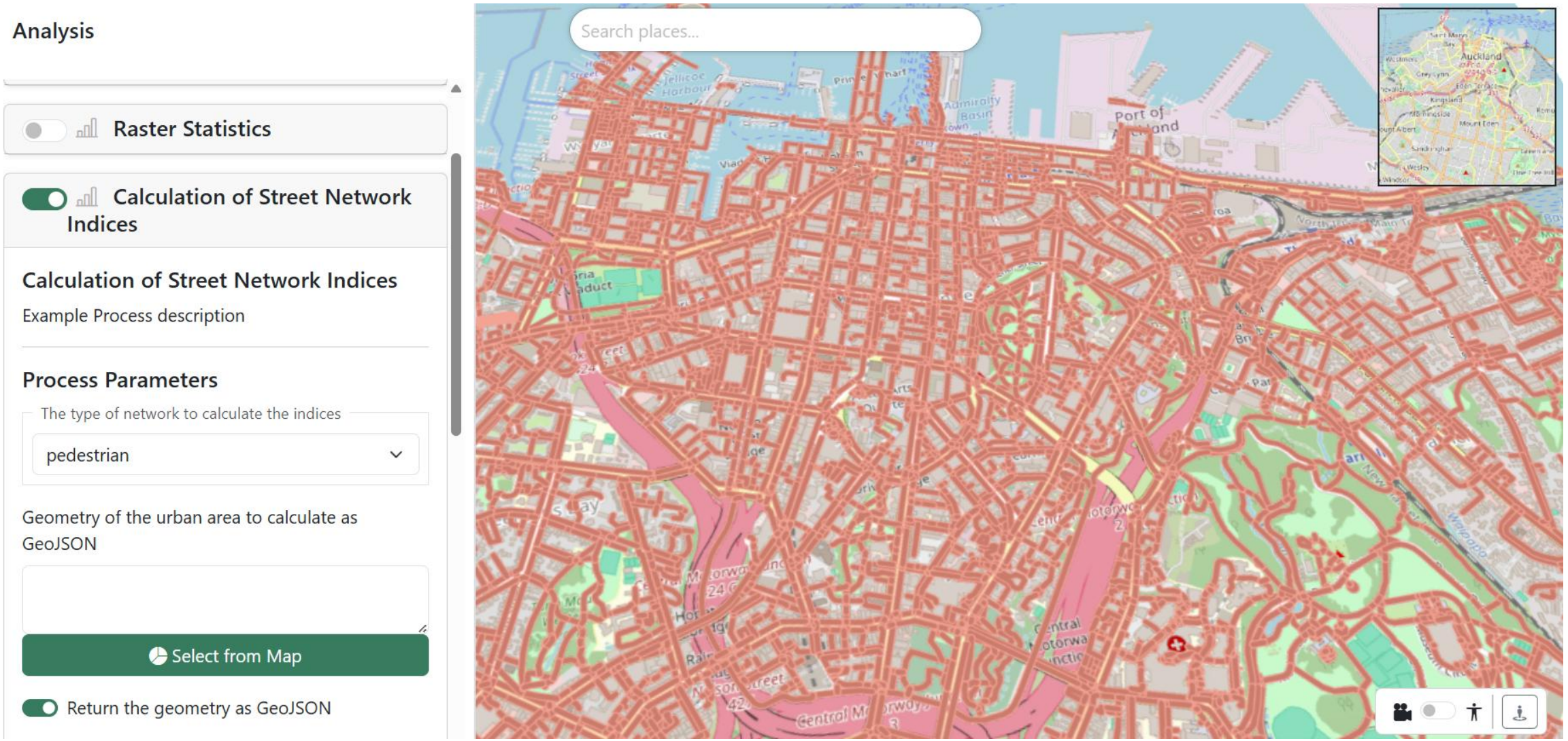
Slope

Circuity

Intersections

Orientation Entropy

Use Case: Urban Digital Twin – Analysis Results



Use Case: Urban Digital Twin – Results

City-scale Street Network analysis for pedestrian mobility

	Pedestrian
Avg. Proximity to POIs	37.3745
Circuity	1.074
Avg. Street Length	51.80
Street Density	14,537
Link-Node Ratio	1.27
Intersection Density	249
Avg. Proximity to Public Transport	8.5547

- **Ideal conditions for pedestrian mobility**
- Low average street lengths for pedestrians.
- Link-Node ratio just over in the range for good connectivity in the pedestrian network.

Conclusions

04

Conclusions

Main achievements:

1. Identified system components and base data for Urban Digital Twins.
2. Development of UrbanDT, an open-source Urban Digital Twin prototyping platform.
3. Formulated a functional automated pipeline for global open data download as default data for UDT prototypes.
4. Implemented an innovative Docker-based automated pipeline for on-demand UDT prototype creation from a bounding box.
5. Demonstrated by implementing a UDT on-the-fly of Auckland, New Zealand, and performing street network analyses.

Conclusions – Future Work

1. Support for monitoring data is still lacking!
2. More standardised formats/interfaces (OGC API – Features, BIM (IFC), CityGML / CityJSON, etc…)
3. Additional support and optimization for visualisation – especially 3D

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GEOLaB Geomatics and Earth Observation lab



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Repository